Fiscal Year:	FY 2024	**Task Last Updated:**	FY 07/20/2023
PI Name:	Sanford, Larry Ph.D.		
Project Title:	Impact of Inflight Stress and Sleep Disturbances on Brain Function, Neural Communication, and Inflammation		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **HFBP**:Human Factors & Behavioral Performance (IRP Rev H)		
Human Research Program Risks:	(1) **BMed**:Risk of Adverse Cognitive or Behavioral Conditions and Psychiatric Disorders (2) **Sensorimotor**:Risk of Altered Sensorimotor/Vestibular Function Impacting Critical Mission Tasks		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	23507-1607	**Congressional District:**	3
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2018 HERO 80JSC018N0001-Crew Health and Performance (FLAGSHIP, OMNIBUS). Appendix A-Flagship, Appendix B-Omnibus
Start Date:	10/01/2019	**End Date:**	09/30/2025
No. of Post Docs:	0	**No. of PhD Degrees:**	1
No. of PhD Candidates:	1	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JSC
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Flight Program:			
Flight Assignment:	NOTE: End date changed to 09/30/2025 per L. Juliette (Ed., 5/8/22)		
Key Personnel Changes/Previous PI:	July 2021 report: Shao-Hui Chuang and Richard Ciavarra are no longer working on the project. July 2020 report: None.		
COI Name (Institution):	Britten, Richard Ph.D. (Eastern Virginia Medical School) Wellman, Laurie Ph.D. (Eastern Virginia Medical School) Yoon, Hargsoon Ph.D. (Norfolk State University)		
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Performance Goal No.:			
Performance Goal Text:			

Task Description:	Astronauts may experience several physical and psychological challenges including prolonged exposure to microgravity, potential social isolation (SI), and predicted exposure to ~13 cGy/Yr Space Radiation (SR). Each of these stressors can have deleterious effects on physical and mental health as well as negatively impact mission performance. Each also can disturb sleep, which can further exacerbate the effects of stress. Determining how astronauts respond to the single and compound effects of mission related stressors is crucial for understanding and mitigating factors that could impede mission success. However, examining their effects will be complex because they have different time courses and durations. SR exposure will be episodic whereas microgravity will be constant and SI and sleep disturbances can be temporary or prolonged. Their effects may also be additive or synergistic, and may vary with sex and with individual differences in stress resilience and vulnerability. Therefore, our goal in this project is to determine the single and synergistic effects of inflight stress and SR on mission related biological and functional challenges that astronauts may face. We will determine how order of experience impacts the effects of inflight stressors and determine sex and individual differences in stress resiliency using an established Wistar rat model. We will use established ground-based models of microgravity (hindlimb unloading), SI, and SR. We will use telemetry to record sleep and to assess temperature increases as an index of the stress response. We will use plasma levels of brain derived neurotropic factor to predict individual differences in stress responses. We will assess fear learning and extinction to determine how chronic inflight stressors impact ability to respond to an additional acute stressor. We will analyze food entrained, anticipatory activity to assess potential impairments in temporal perception. We will determine the effects of inflight stress and SR on sensorimotor function and recognition memory using behavioral tests. We will use local field potential recordings in the amygdala, medial prefrontal cortex, and hippocampus to determine whether stressors and SR alter the coordinated neural activity underlying fear learning and extinction, and we will determine whether there are structural changes within connections between regions. We will use chemokine/cytokine arrays and Ingenuity Pathway Analysis to determine the effect of stressors and SR on the immune system. Collectively these studies will assess functional, mechanistic, and immune consequences of several stressors inherent in space exploration. The data they provide will aid in risk assessment, and may help to develop countermeasures based on individual differences in ability to withstand stress and/or targeting immune pathways that may confer protection against the deleterious effects of multiple stressors that astronauts will face.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Our studies will increase understanding of how individual differences in resilience and vulnerability factor in determining lasting outcomes of stressful experiences. They will also determine the role that stress-induced alterations in sleep play in mediating the outcomes of stress. The results will have importance for understanding how individuals cope with stress on Earth and on missions in space.
Task Progress:	Our goal in this project is to determine the impact and potential synergistic effects that multiple inflight stressors and exposure to space radiation (SR) can have on biological systems and mission related performance. Outcomes that we are examining include sensorimotor function and balance (adhesive removal task and balance beam performance), recognition memory (novel object recognition task), time perception (anticipatory activity), fear memory, and fear memory related neural communication. We are also examining the relationship of altered performance to the effects of inflight stressors on sleep, and how responses to stressors vary in resilient (Res) animals that maintain sleep after experiencing stress and vulnerable (Vul) animals that have reduced sleep after experiencing stress. This is the fourth year of this project. We have now conducted studies in controls (males: n=20; females, n=16), social isolation (social isolation/SI; males: n=21; females: 16), SR (males: n=15; females: n=15), and SI + SR animals (males: n=15; females: n=15). Rats were divided into Res and Vul phenotypes based on rapid eye movement (REM) sleep responses to footshock stress. Some analyses and comparisons across groups are still ongoing but when completed will provide data for comparisons of males and females across inflight stressors. We will also be able to determine whether both males and females show Vul and Res phenotypes. We also are currently pilot testing hindlimb unloading (HLU) in females and will begin sleep and behavioral studies in the current project period. In general, the data we have currently available for sex comparisons demonstrates that SI can produce significant alterations in sleep that differ in males in females. SI females showed significantly greater REM electroencephalogram (EEG) alpha power than control females in the light period. Elevated EEG spectral power in the high frequency range (alpha, sigma, and beta) is considered an indicator of cortical hyperarousal whereas low power in low-frequency bands (delta and theta) may indicate less sleep propensity and quality. Together, our data indicate that SI reduces sleep amount and quality, with differences in males and females that suggest more disrupted sleep in females. We will need to determine whether these differences hold when we have the full data set analyzed as well as the effects of SR and SR + SI on sleep across sexes. We also found that SR exposure had a major negative impact on the integrity of brain macro and micro structures. The altered morphology induced by SR appeared to disrupt both vascular and lymphatic systems, observed by increased BBB permeability and LV enlargement and associated with a striking loss of supporting cells, including astrocytes and endothelial cells. Unexpectedly, the increased permeability in the brain was associated with a suppressed immune response in the SR group and was found to be a result of depleted infiltrating immune cells, despite significant up-regulation of gene expression related to cytokine signaling and immune cell recruitment. Interestingly, some of the deficits observed in the SR group were marginally rescued in the DFS group. Despite this, compared to Control and SI groups, DFS animals still had significant reductions in structural and neuroimmune integrity. Further studies are required to understand the specific molecular changes that lead to these structural alterations, though Nanostring® results and previous literature allude to a loss of important tight junction and matrix-related proteins as potential causation. These alterations in the CNS and neuroimmune responses could account for the increased anxiety and sensorimotor impairments observed in the SR and DFS groups as immune system dysregulation has been shown to increase anxiety and neuronal damage. Additionally, these changes could be a correlation, or a result, of the disturbed sleep also present in these groups, as immune dysregulation is also linked to disturbed sleep, which can also increase neuroinflammation and lead to neuronal damage. A down-regulation of pathways related to neuronal integrity and neurotransmission in these groups was found and could likely be a cause of the learning impairments observed in these groups. Future studies will be necessary to explain specific changes to neuronal viability and neurotransmission. The SI group (males) did not exhibit major differences in structural morphology or immune infiltrate compared to the

Control group. Therefore, SR appears to have a main negative effect on cell viability and structural integrity in the brain while SI may have potential protective properties against the negative effects of SR. However, the synergistic relationship between these two stressors and how they interact is not understood. SI animals did exhibit differences in neuroinflammatory-related gene expression and altered astrocyte morphology compared to control groups. These changes were also found in the DFS group. It has been shown that hyperactive astrocytes can alter memory formation. SI animals did experience blunted learning in sensorimotor tasks, but further studies are required to assess the activation level of these astrocytes within these groups, and how these changes are related to sleep.

Alterations in BBB integrity and neuroimmune signaling were also associated with differences in post-stress sleep, as Vul animals had increased BBB damage and heightened immune response compared to Res animals. The adverse effect of stress may include stress-induced “priming” or sensitization of the neuroinflammatory response that can increase vulnerability to subsequent pro-inflammatory challenges. Previous studies by our lab and others have shown that the stress-induced neuroinflammatory response can be modulated by stress perception. Thus, while stress can promote inflammation, it also can be mitigated by stress resilience and vulnerability that may be regulated by differences in neural circuits that also regulate sleep. Therefore, the stressors that astronauts will likely face can produce deleterious effects both via neuroinflammation and disturbed sleep, and vulnerable individuals may be more likely to develop physical and cognitive deficits associated with these disruptions. Therefore, it is plausible that sleep itself can promote the integrity of the BBB in response to inflight stressors. However, the potential synergistic effects of multiple stressors on the relative changes in the immune system could alter these responses.

In summary, our data to date demonstrate both single and synergistic effects of SI and SR on several behaviors and sleep and also demonstrate differential effects in male Res and Vul rats as well as significant sex differences. Future work will begin to determine how extensive these differences are, and whether Res and Vul phenotypes can be identified in females.

Bibliography Type:	Description: (Last Updated: 03/16/2026)
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